

What Can Conformal Risk Control Certify for PET/CT Tumour Segmentation?

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Abstract

A conformal guarantee for segmentation is useful only when the controlled loss reflects the clinical decision. In PET/CT tumour imaging, missing a whole lesion can matter more than small boundary errors. We apply connected-component risk-controlling prediction sets (RCPS) to fixed, pretrained LesionTracer probability maps for 900 autoPET cases, asking whether one global threshold can certify a binary missed-lesion loss and a continuous voxel-level missed-tumour loss. At $\alpha = \delta = 0.1$, the binary loss cannot be certified: on lesion-positive calibration cases its empirical risk is 0.177 even at the smallest threshold we evaluated. The voxel loss behaves differently: an empirical-Bernstein bound certifies $\hat{\lambda} = 0.78$, giving $\mathbb{P}(R^{\text{vox}}(\hat{\lambda}) \leq 0.1) \geq 0.9$, once lesion-free cases contribute zero loss. This binary infeasibility holds for coverage requirements γ from 0.5 to 0.9. We read the contrast as a statement about the prediction family: a global threshold cannot recover lesions the network never supported.

Keywords: risk-controlling prediction sets; conformal risk control; PET/CT; tumour segmentation; lesion-level uncertainty

1. Problem and Method

In PET/CT tumour segmentation, the clinically relevant error is often lesion-level, while conformal segmentation methods commonly target voxel- or mask-level risk. Building on distribution-free uncertainty quantification (Angelopoulos and Bates, 2021), conformal risk control (Angelopoulos et al., 2022; Bates et al., 2021), and recent morphology-, spatial-, lesion-, and semantic-aware segmentation methods (Mossina et al., 2025; Bereska et al., 2025; Tan et al., 2025; Teneggi et al., 2025), we use RCPS as a diagnostic framework: with the segmentation model fixed, can a single global threshold certify the intended clinical risk?

For a probability map $\hat{p}_i$ of case X_i , the threshold λ defines the nested prediction region

$$T_\lambda(X_i) = \{v : \hat{p}_i(v) \geq \lambda\}, \quad (1)$$

where v indexes voxels; smaller λ gives a larger, more sensitive region. Equation 1 is the standard superlevel-set form of threshold segmentation, not itself a conformal guarantee. For ground-truth lesion component Y_{ik} , coverage and the corresponding binary loss are

$$c_{ik}(\lambda) = \frac{|Y_{ik} \cap T_\lambda(X_i)|}{|Y_{ik}|}, L_i(\lambda) = K_i^{-1} \sum_k \mathbb{I}\{c_{ik}(\lambda) < \gamma\}, \quad (2)$$

where i indexes cases, k the K_i lesions in case i , and γ is the required coverage. Here $c_{ik}(\lambda) \in [0, 1]$ is the fraction of lesion k covered by the region, so Eq. 2 counts a lesion as missed when its coverage falls below the clinically chosen tolerance γ . We contrast it with the continuous voxel loss

$$L_i^{\text{vox}}(\lambda) = \frac{|Y_i \setminus T_\lambda(X_i)|}{|Y_i|}, \quad (3)$$

Table 1: Test-set performance on 245 lesion-positive cases. Recall is lesion recall; FN frac. and FP frac. are false-negative and false-positive voxel fractions. The certified row gives a voxel-risk guarantee; the low-threshold row is non-certifying.

Method	λ	Recall	FN frac.	Prec.	Dice	FP frac.
Standard mask	0.50	0.647	0.105	0.573	0.567	0.092
Fixed relaxed	0.30	0.682	0.095	0.555	0.556	0.106
Fixed relaxed	0.10	0.715	0.080	0.532	0.543	0.129
Binary low-threshold	10^{-5}	0.819	0.046	0.449	0.482	0.230
Voxel RCPS	0.78	–	$\leq \mathbf{0.10}^\dagger$	–	–	–

$^\dagger \mathbb{P}(R^{\text{vox}}(\lambda) \leq 0.1) \geq 0.9$; Bernstein UCB = 0.093.

where $Y_i = \cup_k Y_{ik}$; this study-specific normalised loss measures the omitted tumour fraction and gives partial credit for partial coverage. RCPS selects

$$\hat{\lambda} = \sup\{\lambda : \hat{R}^+(\lambda) \leq \alpha\}, \quad (4)$$

where $\hat{R}^+$ is a finite-sample upper confidence bound (UCB) on the true risk $R(\lambda) = \mathbb{E}[L_i(\lambda)]$ (and R^{vox} for the voxel loss), α is the target risk, and δ is the allowed failure probability. Equation 4 is the standard RCPS/conformal-risk-control selection rule (Angelopoulos et al., 2022; Bates et al., 2021): choose the largest region threshold whose certified upper risk bound is below α . We use Hoeffding for the binary loss and empirical Bernstein (Maurer and Pontil, 2009) for the continuous loss; calibration and test cases are separated. In summary, Eqs. 1–3 specify the prediction family and clinically motivated losses introduced for this diagnostic study, whereas Eq. 4 supplies the published conformal-risk-control decision rule.

2. Evaluation and Results

We use fixed, pretrained LesionTracer (Rokuss et al., 2024) ensemble probability maps on 900 open autoPET cases (Gatidis et al., 2022, 2024), retaining one examination per patient and splitting at patient level into 450 calibration patients and 450 test patients. These cases are held out from the model’s training data, so calibration and test remain exchangeable under the usual i.i.d. patient-sampling assumption. We compare fixed thresholds 0.50, 0.30, 0.10, 10^{-5} ; 10^{-5} is the smallest threshold we evaluated, not a mathematical lower limit. Only RCPS selects a threshold from calibration labels, and test labels are never used for selection. We set $\alpha = \delta = 0.1$ and $\gamma = 0.8$. Calibration risks are patient-averaged; the binary loss uses 230 lesion-positive cases, while the voxel loss uses all 450 cases and assigns zero loss to lesion-free cases.

The binary lesion loss has no feasible RCPS threshold. We report $\lambda = 10^{-5}$ only as a smallest-evaluated-threshold stress test, not as a certified choice. Even there, the empirical calibration risk is $0.177 > 0.1$, before the Hoeffding correction of 0.071; the empirical Bernstein UCB is still 0.242. This is a support failure of the fixed prediction family, not of conformal risk control. The conclusion is robust for $\gamma \in \{0.5, 0.6, 0.7, 0.8, 0.9\}$, whose smallest-evaluated-threshold empirical risks are 0.106, 0.118, 0.139, 0.177, 0.310.

By contrast, the voxel-level empirical risk at $\lambda = 0.5$ is 0.058, and the Bernstein UCB remains below 0.1 through $\hat{\lambda} = 0.78$, where it is 0.093. The held-out test risk at this threshold is 0.067. Relaxing the threshold improves lesion recall but increases false-positive burden, illustrating why a loss that credits partial coverage can be certified even when strict lesion-level certification cannot.

3. Conclusion and Research Direction

The experiment shows that conformal certification is determined jointly by the target risk and expressive power of the prediction family. A global threshold cannot recover lesions that receive insufficient probability support, even when it is made very permissive. The voxel guarantee should therefore not be interpreted as lesion-level certification. The next stage is a lesion-adaptive conformal family that calibrates component-specific regions from PET/CT evidence and jointly controls missed-lesion risk and false-positive burden.

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